SMART INDIA HACKATHON 2026

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TITLE PAGE

- **Problem Statement ID- SIH26016**
- **Problem Statement Title- Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support**
- **Theme- Miscellaneous**
- **PS Category- Software**
- **Team ID- 31**
- **Team Name (Registered on portal)- Code-o Patronum**

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IDEA TITLE

- **Proposed solution:**
  - An end-to-end web app digitizing the RFCTLARR Act 2013 lifecycle, merging a strict legal state-machine workflow with dynamic GIS visualization.
  - Features a mobile-responsive Progressive Web App (PWA) for field officials to geo-tag land offline and sync later.
  - Centralized executive dashboard for real-time tracking of compensation, R&R (Rehabilitation & Resettlement), and project milestones.

- **How it addresses the problem:**
  - **Replaces Fragmented Systems:** Unifies Central, State, and District workflows into one API-driven hub.
  - **Eliminates Opacity:** Role-Based Access Control (RBAC) ensures transparency; citizens track their specific ULPIN (Bhu-Aadhaar) status.
  - **Speeds Up Approvals:** Automated SLA (Service Level Agreement) alerts prevent files from lingering at any desk.

- **Innovation and uniqueness of the solution:**
  - **Bhashini AI Integration:** Automatically translates legal notifications and SMS alerts into the landowner's regional language.
  - **Cryptographic Audit Trails:** Final award documents (PDFs) are hashed (SHA-256) and logged, making post-approval tampering impossible without detection.

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TECHNICAL APPROACH

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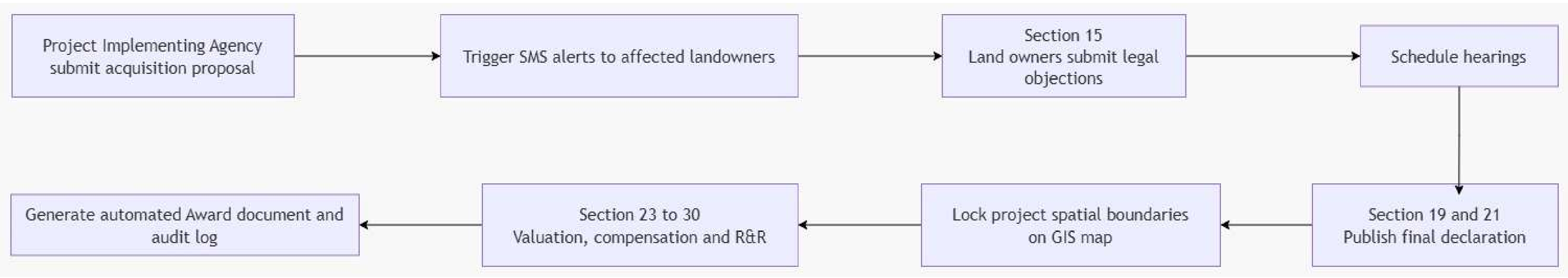
- **Technologies to be used:**

Frontend (Web & Mobile): Next.js (React), Tailwind CSS, PWA (for offline field sync).
Backend & APIs: FastAPI (Python) for high-performance processing and automated workflows.
Database & Spatial: PostgreSQL with PostGIS extension (Core Data), GeoServer (Tile serving).
Mapping UI: MapLibre GL JS (for rendering heavy cadastral vector boundaries).
Cloud & Storage: AWS S3 (Document repository), Docker (Containerization).

- **Methodology and process for implementation:**



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FEASIBILITY AND VIABILITY

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- **Analysis of the feasibility of the idea:**
  - The stack relies entirely on open-source technologies (PostGIS, React, Python) widely adopted by modern government DPIs (Digital Public Infrastructure).
  - Interoperable: Designed API-first to share information with existing databases like PM GatiShakti and Bhoomi Rashi.

- **Potential challenges and risks:**
  - **Poor Rural Connectivity:** Workers in remote areas might not have the internet connection needed to upload location data from the field.
  - **Legacy Data Quality:** Existing state land records (cadastral maps) may have overlapping or inaccurate polygon boundaries.
  - **Document Tampering:** There is a risk that bad actors might try to secretly change payment or compensation documents after they have already been approved.

- **Strategies for overcoming these challenges:**
  - **Offline-First Architecture:** The React PWA utilizes local browser storage to cache geo-tagged field entries, syncing automatically upon network reconnection.
  - **Spatial Validation:** Backend logic flags boundary overlaps during the proposal phase for manual district collector review.
  - **Zero-Trust Security:** Document hashing ensures any modified file fails the system's automated integrity check.

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IMPACT AND BENEFITS

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- **Potential impact on the target audience:**
    - **For Citizens/Farmers:** It gives landowners easy updates on their mobile phones in their own language. They can quickly check their hearing dates, how their payment is calculated, and the status of their relief and resettlement. This helps reduce worry and cuts down on legal fights.
    - **For District Collectors (CALA):** Replaces physical file-chasing with a prioritized digital inbox and automated calculation multipliers for rural land.

- **Benefits of the solution:**
    - **Economic:** Speeds up the building of big national projects (like highways and railways) by cutting down on waiting time and saving money lost during delays.
    - **Social:** Makes sure that families who have to move get their relief and resettlement money fairly, on time, and in a way that can be easily tracked.
    - **Governance:** Gives the Central Government a complete, live view of projects across all states through an easy-to-use dashboard, helping them see exactly where things are stuck or delayed.

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RESEARCH AND REFERENCES

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- **Details / Links of the reference and research work:**
  - The Right to Fair Compensation & Transparency in Land Acquisition, RR Act, 2013 (Sections 11, 15, 19-30)
  - DILRMP - Digital India Land Records Modernization Programme (dilrmp.gov.in)
  - Unique Land Parcel Identification Number (ULPIN) / Bhu-Aadhaar implementation guidelines by the Department of Land Resources (DoLR)